# Wawatahia University Database Design Assignment

## 1. Introduction

This assignment presents the design and implementation of a database system for Wawatahia University. The system addresses inefficiencies in managing student records and project supervision.

## 2. Planning

The objective is to design a structured database to store and manage student, professor, course, project, and faculty data efficiently.

## 3. System Analysis

Key entities identified: Student, Professor, Course, Project, Faculty. The system must support relationships such as enrollments, teaching, and project supervision.

## 4. Conceptual Design (ER Overview)

Entities:
- Student (NSN, Name, Email)
- Professor (ID, Name, Phone, Email)
- Course (ID, Name)
- Project (ID, Name, Dates, Budget)
- Faculty (ID, Name, HoF)

Relationships:
- Students enroll in courses (many-to-many)
- Professors teach courses (many-to-many)
- Students work on projects (one-to-many)
- Professors assess projects (one-to-many)
- Faculties contain students, professors, and courses

## 5. Logical Design (Tables)

Faculty(FacultyID, FacultyName, HoFName, NumStudents)
Student(NSN, Name, Email, FacultyID)
Professor(ProfessorID, Name, Phone, Email, FacultyID)
Course(CourseID, CourseName, FacultyID)
Project(ProjectID, ProjectName, StartDate, EndDate, Description, Budget, ProfessorID)
Enrollment(NSN, CourseID)
Teaching(ProfessorID, CourseID)

StudentProject(NSN, ProjectID)

## 6. Normalization

The database is normalized to Third Normal Form (3NF), ensuring no redundancy and maintaining data integrity.

## 7. Implementation (SQL Example)

CREATE TABLE Student (NSN INT PRIMARY KEY, Name VARCHAR(100), Email VARCHAR(100));
CREATE TABLE Professor (ProfessorID INT PRIMARY KEY, Name VARCHAR(100));
... (other tables implemented similarly)

## 8. Testing

Sample queries verify relationships such as retrieving students and their assigned projects.

## 9. Evaluation

The new system improves efficiency, scalability, and data consistency compared to the old system.